PRATICAL-9

AIM: Create the Data staging area for the selected database using SQL

STEPS:-

Step 1:- Create a Staging Database

First, create a staging database to store raw sales data.

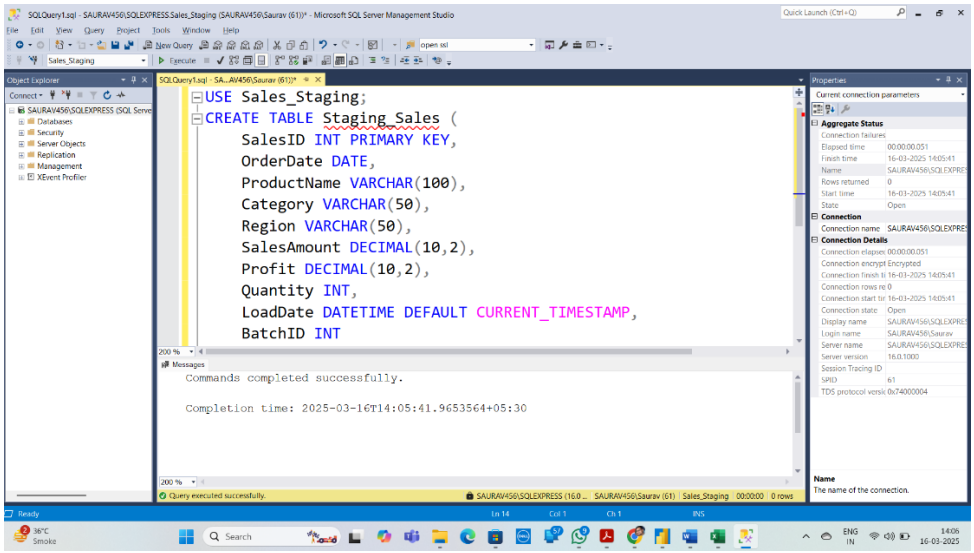
```
CREATE DATABASE Sales_Staging;

USE Sales_Staging;
```

Step 2:- Create Staging Tables

Create tables that match the structure of raw sales data but include additional fields like **load date** and **batch ID**.

```
CREATE TABLE Staging_Sales (
    SalesID INT PRIMARY KEY,
    OrderDate DATE,
    ProductName VARCHAR(100),
    Category VARCHAR(50),
    Region VARCHAR(50),
    SalesAmount DECIMAL(10,2),
    Profit DECIMAL(10,2),
    Quantity INT,
    LoadDate DATETIME DEFAULT CURRENT_TIMESTAMP,
    BatchID INT );
```



Step 3:- Load Raw Data into the Staging Table

Simulating data load from a **CSV file, API, or external source**:

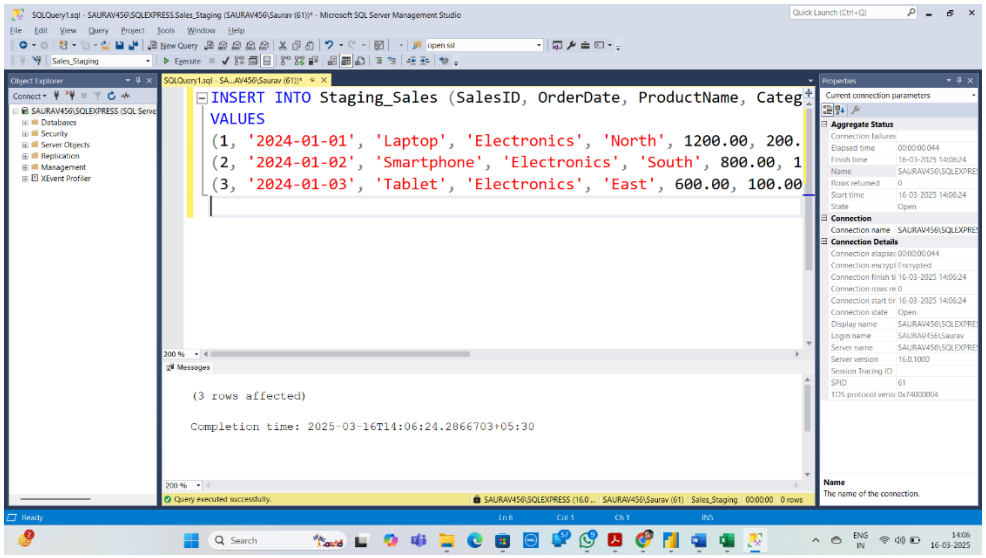
INSERT INTO Staging_Sales (SalesID, OrderDate, ProductName, Category, Region, SalesAmount, Profit, Quantity, BatchID)

VALUES

(1, '2024-01-01', 'Laptop', 'Electronics', 'North', 1200.00, 200.00, 3, 101),

(2, '2024-01-02', 'Smartphone', 'Electronics', 'South', 800.00, 150.00, 2, 101),

(3, '2024-01-03', 'Tablet', 'Electronics', 'East', 600.00, 100.00, 5, 101);



Step 4:- Perform Data Cleansing & Transformation

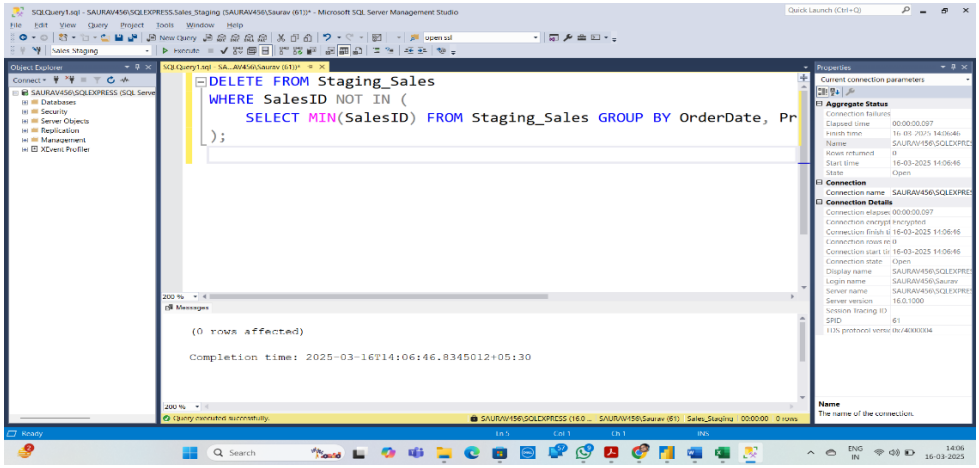
- **Remove Duplicates**

  DELETE FROM Staging_Sales

  WHERE SalesID NOT IN (

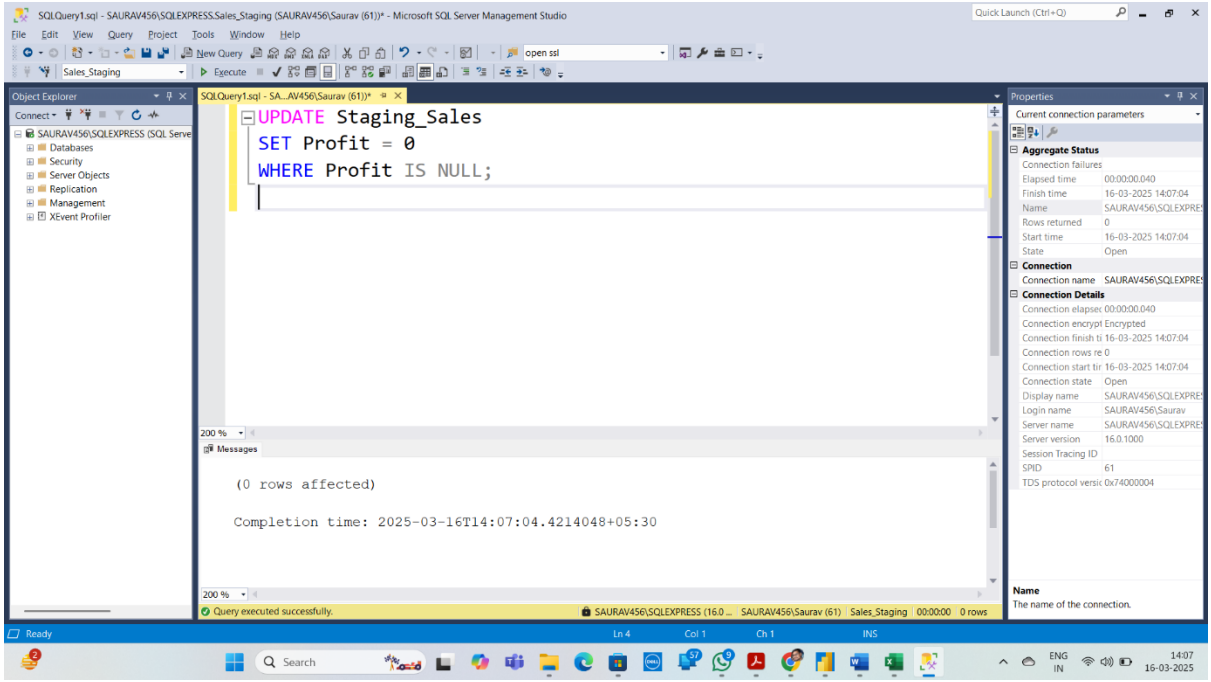
  SELECT MIN(SalesID) FROM Staging_Sales GROUP BY OrderDate, ProductName, Region

  );



- **Handle Null Values**

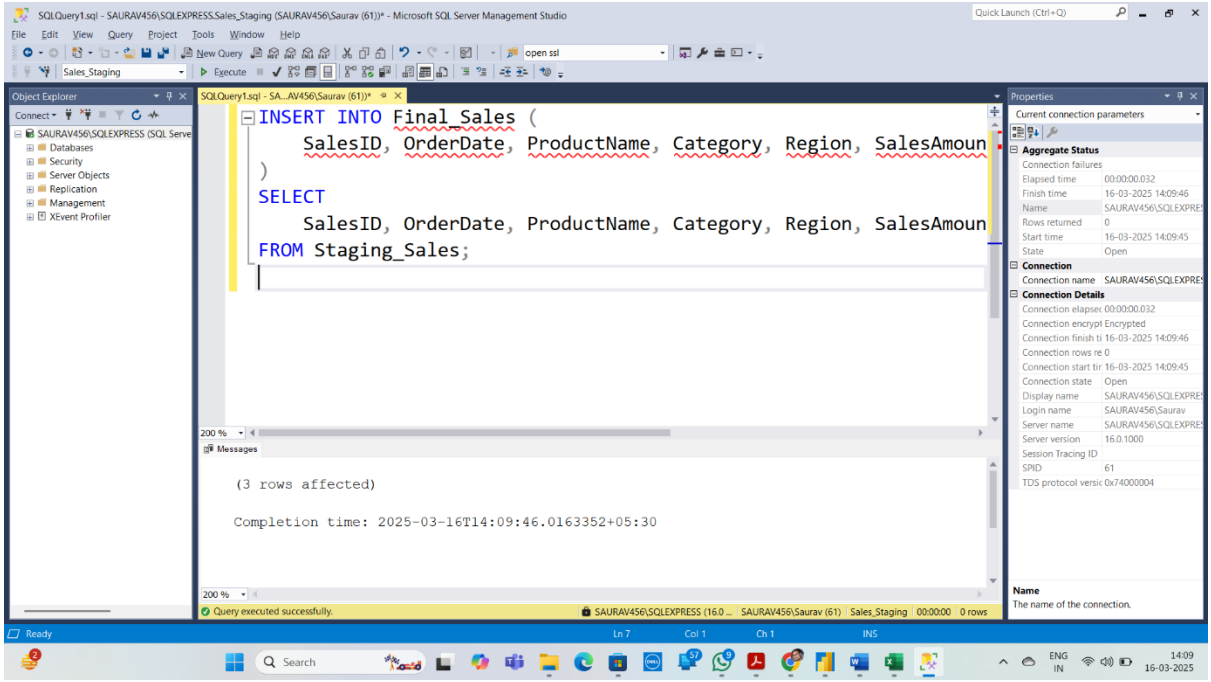
```
UPDATE Staging_Sales
SET Profit = 0
WHERE Profit IS NULL;
```



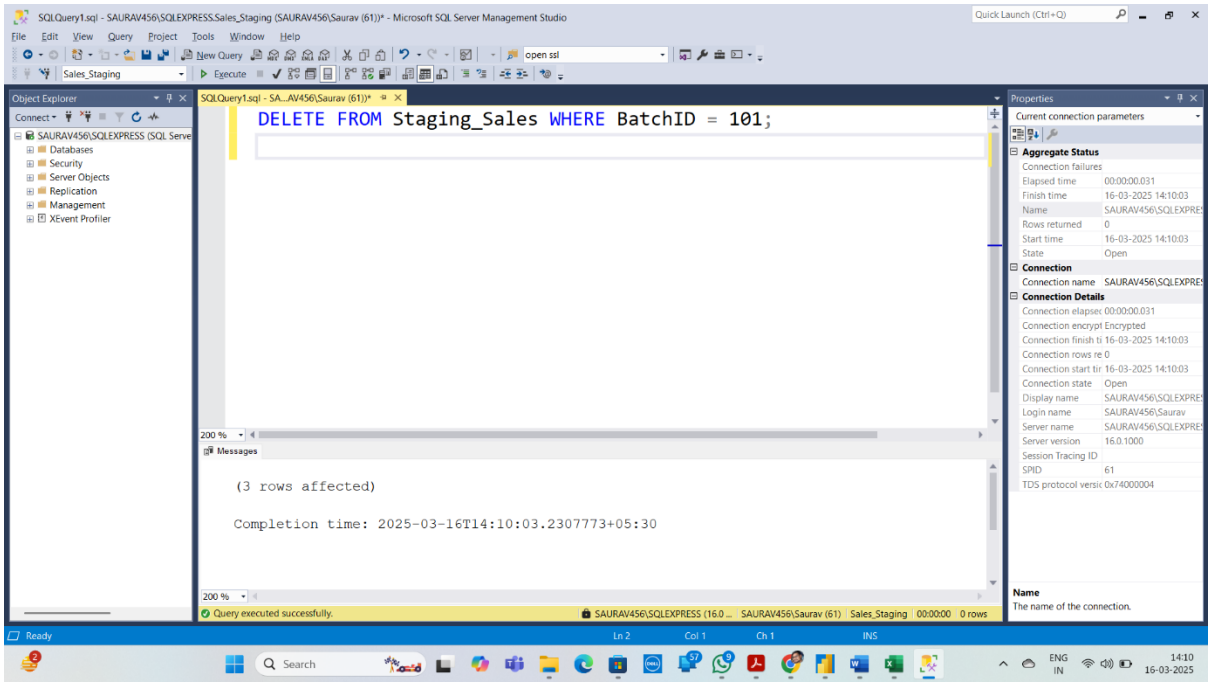
<u>Step 5:-</u> Transfer Clean Data to the Final Sales Table

Move the cleaned data into the Data Warehouse (DWH).

```
INSERT INTO Final_Sales (
   SalesID, OrderDate, ProductName, Category, Region, SalesAmount, Profit, Quantity
)
SELECT
   SalesID, OrderDate, ProductName, Category, Region, SalesAmount, Profit, Quantity
FROM Staging_Sales;
```



Step 6:- Archive or Delete Processed Data

Once data is loaded, either archive it or delete it from the staging area.

DELETE FROM Staging_Sales WHERE BatchID = 101;